

Medical Insurance Cost Prediction

Project Story Report

Chapter 1: The Business Problem

Medical insurance companies need to estimate customer medical charges accurately. Incorrect estimates can lead to financial losses, poor pricing decisions, and reduced customer satisfaction. The objective of this project was to build a machine learning model capable of predicting insurance charges using historical customer information.

Chapter 2: Understanding the Data

The dataset contains 1,338 customer records with demographic and lifestyle information including age, sex, BMI, number of children, smoking status, and region. The target variable is the medical insurance charge. Initial exploration confirmed there were no missing values and only one duplicate record.

Chapter 3: Preparing the Data

After exploring the data, duplicate records were removed. Categorical variables were converted into numerical values using encoding techniques, making the dataset suitable for machine learning algorithms. Finally, the data was split into training and testing datasets to evaluate how well the models generalize.

Chapter 4: Searching for the Best Model

Rather than assuming one algorithm would perform best, four regression models were evaluated: Multiple Linear Regression, Support Vector Regression, Decision Tree Regressor, and Random Forest Regressor. Hyperparameter tuning was applied where appropriate to improve model performance.

Chapter 5: The Results

Each model was evaluated using the R^2 score. Multiple Linear Regression achieved 0.7709, Decision Tree achieved 0.6875, Support Vector Regression achieved 0.5020, while Random Forest Regressor achieved the highest score of 0.8198.

Chapter 6: Why Random Forest Won

Random Forest Regressor consistently explained about 82% of the variation in insurance charges, outperforming every other model. Unlike linear regression, it captures complex non-linear relationships and interactions between variables such as age, BMI, and smoking status. It also reduces overfitting by combining predictions from many decision trees, making it more reliable for unseen data.

Business Impact

A reliable prediction model enables insurers to estimate expected medical costs more accurately, support premium pricing, identify high-risk customers, and make faster data-driven decisions.

Final Recommendation

Random Forest Regressor is recommended for deployment because it achieved the highest R^2 score (0.8198), demonstrated better predictive capability than the other evaluated models, and is robust against overfitting due to its ensemble learning approach. If interpretability becomes the primary business requirement, Multiple Linear Regression can be retained as a simpler baseline; however, for predictive performance, Random Forest is the strongest choice.